



利民化学有限责任公司 LIMIN CHEMICAL CO., LTD.

Address: Economic Development Zone, Xinyi, Jiangsu, 221400, China
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MATERIAL SAFETY DATA SHEET

FOSETYL-ALUMINIUM 80% WDG

SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name: Fosetyl -Aluminium 80% WDG

Use: Fungicide

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SECTION 2 - COMPOSITION, INFORMATION ON INGREDIENTS

Active Ingredient: Fosetyl -Aluminium

CAS#: 39148-24-8

Chemical Family: Aluminum ester of alkyl phosphonate

Chemical Formula: $C_6H_{18}AlO_9P_3$

IUPAC: aluminium tris-O-ethylphosphonate

Chemical Name	CAS#	Proportion(g/kg)
Fosetyl Al	39148-24-8	800
Other inert		200

SECTION 3 - HAZARDS IDENTIFICATION

Risk phrased: R36-Irritating to eyes

SECTION 4 - FIRST AID MEASURES



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Eye contact: Hold eye open and rinse slowly and gently with water for 15-20 minutes.

Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice

Skin contact or clothing: take off contaminated clothing. rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: if inhaled remove to fresh air and keep at rest. obtain medical advice if symptoms persist. If not breathing give artificial respiration and get medical attention as soon as possible.

Ingestion: wash out mouth with water. Do not induce vomiting. Keep patient at rest and seek medical advice as above.

SECTION 5 - FIRE FIGHTING MEASURES

Extinguishing media: Foam, carbon dioxide, dry powder

Hazards from combustion products: In a fire, oxides of carbon, nitrogen and phosphorus gas may be formed.

Precautions for fire fighters: dust may form explosive with air. Toxic decomposition products may be produced in a fire. Fire fighters should wear full protective gear, including self-contained breathing apparatus. keep unnecessary people away. Bund area with sand or earth to prevent contamination of drains or waterways. Dispose of extinguishing agent and spillage safely later.

SECTION 6 - ACCIDENTAL RELEASE MEASURES

Evacuation Procedures and Safety: Wear appropriate protective gear for the situation. See Personal Protection information in Section 8. Ventilate closed spaces before entering. Stay upwind if possible.



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Containment of Spill: Follow procedure described below under Cleanup and Disposal of Spill.

Cleanup and Disposal of Spill: Avoid creation of dusty conditions. Scrape up and place in appropriate closed container (See Section 7: Handling and Storage). Decontaminate tools and equipment following cleanup.

Environmental and Regulatory Reporting: If spilled on the ground, the affected area should be scrapped clean and placed in a appropriate container for disposal. Do not flush to drain. Prevent material from entering public sewer system or any waterways.

SECTION 7 - HANDLING and STORAGE

Handling: keep out of reach of children. May irritate the eyes. avoid contact with eyes and skin. Avoid inhaling dust. wash hands after use.

Storage: store in the closed, original container in a dry, cool, well-ventilated area, out of direct sunlight.

Flammability: low flammability. however, if a dust forms, dust/air mixtures can be explosive.

SECTION 8 - EXPOSURE CONTROLS, PERSONAL PROTECTION

Exposure standards:

The NOHSC exposure standard (TWA) for precipitated silica is 10 mg/m³.

The ACGIH TLV – TWA for Aluminium compounds, expressed as aluminium is 2 mg/m³.

Definitions:

Exposure standard – Time Weighted Average (TWA) means the average airborne concentration of a particular substance when calculated over a normal eight-hour working day, for a five-day working week.



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TLV – TWA = Threshold Limit Value – Time Weighted Average. The TWA concentration for a conventional 8-hour workday and a 40-hour work week, to which it is believed that nearly all workers may be repeatedly exposed, day after day, for a working lifetime without adverse effect.

ACGIH = American Conference of Governmental Industrial Hygienists

Biological limit values: None allocated

Engineering controls: Control process conditions to avoid contact. Use local exhaust ventilation to keep exposure levels below the exposure limits above and in situations where dust may be formed.

Personal Protective Equipment:

Eyes: Safety goggles

Clothing: Cotton overalls buttoned to the neck and wrist

Gloves: Elbow-length PVC gloves

Respiratory: If airborne concentrations are likely to exceed the exposure standards

SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES

Appearance: flowable off-white granules.

Odour: acidic odour

Vapor pressure: < 0.013 mPa at 25°C (fosetyl-aluminium)

Volatility: Not available

Solubility in water :dispersible in water

Flash point: not applicable

SECTION 10 - STABILITY AND REACTIVITY

Chemical stability: stable under normal conditions of use.

Conditions to avoid: Heat, moisture. Prevent formation of dust.

Incompatible materials: Avoid strong acids, strong bases, strong oxidising agents, strong



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reducing agents, and extreme humidity.

Hazardous decomposition products: In a fire, oxides of carbon, nitrogen and phosphorus and phosphine gas may be formed.

Hazardous reactions :None

SECTION 11 - TOXICOLOGICAL INFORMATION

POTENTIAL HEALTH EFFECTS

Inhalation This product is expected to have low toxicity by inhalation, but dust may cause respiratory irritation.

Skin contact May irritate skin.

Eye contact Irritating to eyes.

Ingestion Low toxicity, but may be harmful if a large amount is swallowed.

MAMMALIAN TOXICOLOGY

Fosetyl-Aluminium

Oral Acute oral LD₅₀ for rats >7080 mg/kg.

Skin and eye Acute percutaneous LD₅₀ for rats and rabbits >2000 mg/kg. Non-irritating to skin.

Inhalation LC₅₀ (4 h) for rats >5.11 mg/l air.

NOEL NOAEL (2 y) for dogs 298 mg/kg b.w. Non-teratogenic and non-mutagenic.

ADI 2.98 mg/kg b.w (company proposed)

Other Non-carcinogenic.

Toxicity class WHO (a.i.) U; EPA (formulation) III

SECTION 12 - ECOLOGICAL INFORMATION

Fosetyl-Aluminium



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Birds Acute oral LD₅₀ for bobwhite quail >8000 mg/kg. Dietary LC₅₀ (5 d) for bobwhite quail and mallard duck >20 000 ppm diet.

Fish LC₅₀ (96 h) for rainbow trout 428.1, bluegill sunfish 173.2 mg/l.

Daphnia LC₅₀ (48 h) >100 mg/l.

Algae EC₅₀ (90 h) for *Scenedesmus pannonicus* 21.9 mg/l.

Bees LD₅₀ (96 h, oral) >461.8 µg/bee; (contact) >1000 µg/bee

Worms LC₅₀ (14 d) >1000 mg/kg.

SECTION 13 - DISPOSAL CONSIDERATIONS

Waste Disposal Method: Chemical additions, processing, or otherwise altering this material may make the waste management information presented in this MSDS incomplete, inaccurate, or otherwise inappropriate. Please be advised that state and local requirements for waste disposal may be more restrictive or otherwise different from federal laws and regulations. Consult state and local regulations regarding the proper disposal of this material.

EPA Hazardous Waste -- No.

SECTION 14 - TRANSPORT INFORMATION

14.1 Transport ADR/ RID/AND: the product is not subject to ADR/ RID/AND regulation.

14.2 Transport IMDG: the product is not subject to IMDG regulation.

14.3 Transport ICAO-TI / IATA: the product is not subject to ICAO-TI / IATA regulation.

14.4 Other information: No data available

14.5 Environment hazards: information on environmental hazards. If relevant. Please see 14.1-14.3

14.6 Special precautions for user: No data available.

14.7 Transport in bulk according to Annex II of Marpol and IBC code: not relevant

SECTION 15 - REGULATORY INFORMATION



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UNEP Persistent Organic Pollutant(POP): not listed

UNEP Prior informed Consent Chemical(PIC):not listed

WHO Obsolete Pesticide: not listed.

SECTION 16 - ADDITIONAL INFORMATION

The information contained in the Safety Data sheet is correct to the best of our knowledge at the date of issue. It is intended as a guide for the safe use, handling, disposal, storage and transportation and is not intended as a warranty or as a specification. The information relates only to the product specified and may not be suitable for combinations with other materials or in processes other than those specifically described herein.

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